Assignment: Netflix User Analytics

Scenario

Netflix is an online movie and TV streaming platform with thousands of subscribers. The company wants to better understand user behaviour and use Machine Learning to improve customer retention and business decisions.

The management has provided a dataset containing information about **750 users**. The dataset includes user demographics, viewing habits, subscription details, and spending patterns.

Your task is to analyze the data and apply Machine Learning techniques to answer key business questions.

Dataset Description

Column	**Description**
UserID	Unique user identifier
Age	User age
Gender	Male/Female
SubscriptionType	Basic/Premium/VIP
WatchHoursPerWeek	Hours watched per week
DevicesUsed	Number of devices used
FavoriteGenre	Action, Comedy, Drama, Sci-Fi, etc.
AdClicks	Number of advertisement clicks
MonthlySpend	Monthly spending (₹)
SubscriptionRenewed	Yes/No

Tasks

Part A: Dataset Understanding

Q1.

Load the dataset and display the first five records.

Q2.

Determine the number of rows and columns in the dataset.

Q3.

Display all column names.

Q4.

Identify numerical and categorical features.

Q5.

Check whether the dataset contains missing values.

Part B: Exploratory Data Analysis

Q6.

Calculate the average age of users.

Q7.

Determine the average watch hours per week.

Q8.

Find the average monthly spending of users.

Q9.

Count the number of users in each subscription category.

Q10.

Determine the percentage of users who renewed their subscriptions.

Part C: Data Preparation

Q11.

Convert categorical features into numerical form.

Q12.

Define the feature set (X) and target variable (y) for subscription renewal prediction.

Q13.

Split the dataset into training and testing sets.

Part D: Decision Tree Classification

Q14.

Train a Decision Tree model to predict whether a user will renew their subscription.

Q15.

Evaluate the model using accuracy.

Q16.

Generate and interpret the confusion matrix.

Part E: K-Nearest Neighbors (KNN)

Q17.

Train a KNN classifier with K = 5.

Q18.

Compare the accuracy of KNN with the Decision Tree model.

Part F: Linear Regression

Q19.

Train a Linear Regression model to predict monthly spending.

Q20.

Predict the monthly spending for a new user and interpret the result.

Business Reflection Questions

1. Which factors appear to influence subscription renewal the most?
2. Why is subscription renewal a classification problem?
3. Why is monthly spending a regression problem?
4. Which algorithm performed better for renewal prediction?
5. How could the platform use these predictions to improve customer retention?